

Literature Review: Perovskite Semiconductors in Photovoltaic Cells

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Abstract – Perovskite-based photovoltaic cells are showing increasingly impressive power conversion efficiency in lab-based experiments. This paper will outline the state of perovskite photovoltaic cell development and the techniques that have allowed them to reach impressive efficiencies in a relatively short period of time. An overview of the challenges that perovskites face and a focus on areas of research that need to be developed further to mitigate the instabilities present in the material. Future iterations of perovskite-based photovoltaic cells have the possibility of expanding PV adoption by allowing for novel installation opportunities including on curved surfaces and in semi-transparent windows.

Index Terms – Perovskite semiconductors, Photovoltaic cells

I. INTRODUCTION

With the rise in demand for clean energy solutions, research into photovoltaic (PV) technologies has been advancing semiconductor technology to new heights. Recently, perovskite semiconductors have shown increasingly impressive lab-based PV cell efficiencies in a short period compared to other PV technologies. Increasing PV cell efficiency is important to maximize the return on investment and lower the break-even time when investing in PV systems. Combining different PV semiconductors into multijunction cells, researchers have been able to raise the maximum power conversion efficiency (PCE) of Silicon (Si) and perovskite-based cells to ~34%, past the theoretical PCE of Si-based cells [1]. By combining perovskites with existing, better-understood PV technologies like Si, researchers can build on previous breakthroughs in Si-based PV research to create a more

reliable and long-lasting solar cell. In addition to efficiency improvements, perovskite cells open new possibilities for the deployment of PV systems due to their thin layer height which allows for transparent flexible cells [2]. Increasing the surface area of perovskite cells larger than small-scale lab tests (area $<1\text{ cm}^2$) has proven to be an issue that needs addressing due to the degradation in efficiency that comes from increasing cell surface area [3]. Perovskite semiconductors in PV cells also have intrinsic instabilities including rapid degradation of their PCE when exposed to heat and humidity [4]. More research into understanding these degradation methods is required before the technology can be commercially available. Lead (Pb) based perovskites are currently the only category of perovskite based PV cells that can reach high PCE viable for commercial use [4]. Further research and development will need to be directed towards understanding the risks that Pb-based PV panels pose to individuals and the environment. If it is determined that the environmental and human risk of lead-based perovskites is too high for widespread use, the development of a lead-free perovskite with a high PCE will be the focus of future advancements in this technology. Perovskite semiconductors in PV cells have the potential to grow the adoption of PV by allowing for new mounting opportunities including on curved surfaces as well as integration into transparent windows. Further research into perovskite semiconductors in PV cells is required to scale the technology to a commercially available product.

II. PV BASICS

Photovoltaic systems contain modules made with a semiconductor material, typically Si, that can generate electron-hole pairs when exposed to light of a certain wavelength. This is made possible by placing a p-type and n-type doped semiconductor next to each other which creates an electric field which separates the positive and negative charges when they are exposed to light. These charges are then carried away as current by a conductive metal. The wavelengths of light that can generate electron-hole pairs in the semiconductor are dependent on the material properties of the semiconductor. Si requires 1.12 electron volts (eV) of bandgap energy to move an electron to the conduction band, which is where the electron can move freely in the material and be carried away as current [5]. Only around $\frac{3}{4}$ of the Sun's radiation radiates with an energy above 1.12 eV which creates a theoretical limit for the maximum efficiency that a solar cell can achieve using Silicon as the semiconductor [5]. Another limiting factor of solar cell efficiency is that any energy above the bandgap energy required to move an electron to the conduction band will be lost as heat [5]. These efficiency reductions create a theoretical efficiency limit of 33% for single-junction Si PV systems using the solar irradiance received on the surface of the Earth [5]. Crystalline-silicon Photovoltaic cells are currently the most popular PV technology, accounting for over 94% of PV module shipments in 2014 [6, p. 23]. This technology can be classified as either monocrystalline or multicrystalline. Multicrystalline cells are generally cheaper due to their larger ingot size during manufacturing which leads to higher throughput whereas monocrystalline cells are more efficient because of the higher material quality which allows for higher charge carrier lifetimes [6, p. 23]. These cells have matured through many iterations of technology and manufacturing processes to optimize their efficiency for commercial use. With these advancements, crystalline silicon PV

can reach efficiencies of 18%-22% with a lifespan of 25+ years under standard test conditions [7].

III. PEROVSKITE SOLAR CELLS

Perovskite solar cells (PSCs) work similarly to traditional Si solar cells except they utilize a semiconductor material with a perovskite crystal structure. This structure is of the form ABX_3 and can have different elements in each variable which changes the electrical properties of the material, allowing for bandgap steering [8]. Another advantage that PSCs have over traditional PV technologies is that the semiconductor is applied in a thin layer, referred to as thin-film, which allows for thin, flexible cells which can open new possibilities to where PV systems could be installed [9, p. 1]. These flexible cells could be installed on curved surfaces or in places where traditional bracket-mounted panels will not fit. PSCs can also be used to create semi-transparent solar cells [2], which can be combined with traditional glass panes to create a window that is able to generate electricity. The maximum PCE of a single junction PSC using methyl ammonium lead iodide ($MAPbI_3$) has a theoretical maximum PCE of 30.5% which is slightly lower than Si [10]. Research into perovskite semiconductors utilized in PV applications is relatively new, starting in 2005 [9, p. 8]. In this short period of time, PSCs lab power conversion efficiency (PCE) has grown to 26.1%, exceeding traditional thin-film technologies including CdTe and CIGS, and approaching the industry standard crystalline silicon cell technology as seen in [1, Figure 1].

Best Research-Cell Efficiencies

Source: NREL 2023

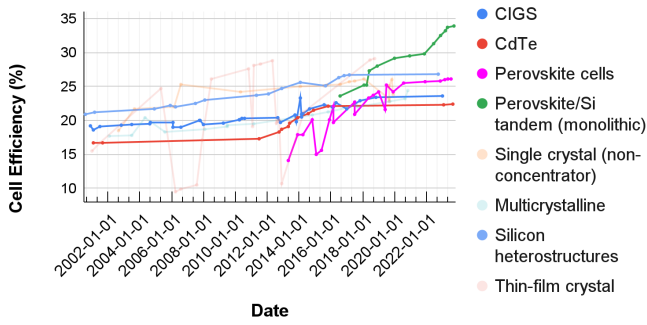


Figure 1. Best Research-Cell Efficiencies

The rapid growth in PSCs efficiency in a short period of time shows its potential for continued research and development. If current efficiency trends continue, PSCs will become a major technology in the global PV market due to the demand for more efficient, compact and versatile solar cell technologies.

IV. MULTIJUNCTION CELLS AND BANDGAP STEERING

Single junction Si solar cells are reaching their theoretical maximum PCE which is leading researchers to develop new methods of exceeding this efficiency [9, p. 434]. One approach is to create a multijunction cell which combines multiple semiconductor materials to collect different parts of the solar spectrum. Perovskites are an ideal candidate for multijunction solar cells because of their high single junction efficiency combined with their chemically tunable bandgap (1.2 eV - 3 eV) [9, p. 435]. Perovskites bandgap tunability allows for researchers to optimize the efficiency of the cell to collect more of the solar spectrum that its partner semiconductor could not collect. Si is a low bandgap semiconductor (1.12 eV) which can be used to efficiently capture the near-infrared (NIR) part of the spectrum [9, p. 436]. Used in a multijunction cell, this layer would be below the perovskite layer which can have a wide bandgap of ~ 1.7 eV [3]. Placing the low bandgap Si under the wide bandgap perovskite allows for the NIR part of the spectrum, which is below the bandgap of the

perovskite and thus does not have the energy required to generate current in the semiconductor, to pass through and be captured by the Si layer. The solar irradiance and spectral absorption of each layer in the solar cell are shown in [11, Figure 2].

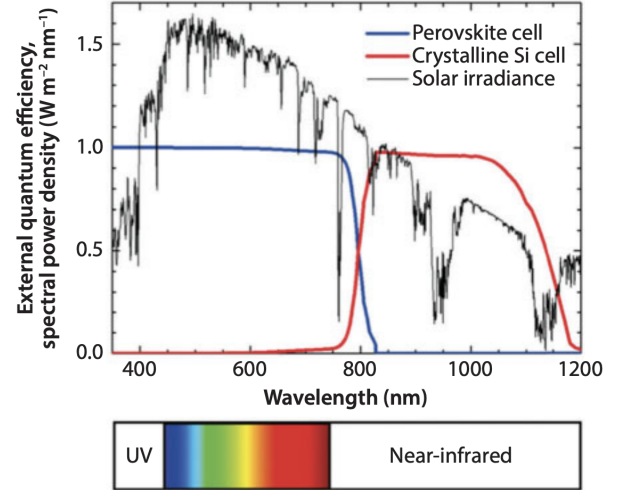


Figure 2. Si/Perovskite Multijunction Cell Spectral Absorption

Utilizing Si in multijunction cells creates a cell with better-understood instabilities due to the immense amount of research and development that has gone into Si PV cells in recent years. Multijunction Si-perovskite cells have been able to exceed the theoretical PCE of single junction Si and perovskite cells, reaching $\sim 34\%$ efficiency as seen in [1, Figure 1]. Multijunction cells utilizing the bandgap steerability of perovskites show a promising future for development and subsequent cost reduction for the technology to be commercialized.

V. STABILITY, SCALABILITY AND ENVIRONMENTAL CONCERNS

Presently PSCs have several problems which need more research to develop the technology into a scalable commercial product. Perovskite solar cells suffer from an intrinsic instability caused by heat which results in degradation of the semiconductor material, lowering its PCE [9, p. 20]. This is important because when a solar cell is illuminated, heat is generated by the energy it receives above the

bandgap energy of the semiconductor material. The degradation caused by heat on the perovskite semiconductor material is substantial with the most efficient PSCs reaching 80% of its original PCE (T_{80}) value in thousands of hours (<1 year) under constant illumination [12]. Existing commercial PV installations degrade 0.8%/year with an expected T_{80} of 25+ years [13]. Although T_{80} values are not close to what will be expected of them when commercialized, research has been done to increase the T_{80} value of PCEs by adding a capping layer of $\text{Cs}_2\text{PbI}_2\text{Cl}_2$ on top of the perovskite to stabilize the semiconductor [12]. Researchers in [12] were able to model accelerated degradation using temperatures as high as 110°C, with a relative humidity (RH) of 65%, and found that the capping layer increased the T_{80} value to >5 years at 35°C. Research focussed on increasing the T_{80} value for PSCs is essential to the commercialization of the technology. In addition to intrinsic instabilities, PSCs also suffer from rapid degradation from humidity in the environment in which they are operating [9, p. 9]. As the perovskite semiconductor is exposed to moisture, its PCE drops dramatically as seen in [4, Figure 3].

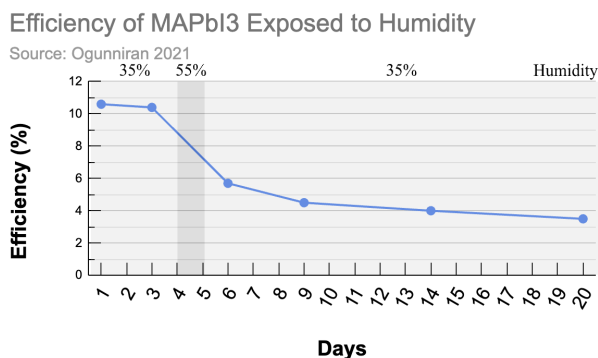


Figure 3. Efficiency of MAPbI₃ Exposed to Humidity

Researchers in [12] have shown that adding the capping layer of $\text{Cs}_2\text{PbI}_2\text{Cl}_2$ on top of the perovskite not only stabilizes the semiconductor's intrinsic stabilities from temperature increases but also mitigates the effect that RH has on the PCE of a PSC.

Scaling up perovskite's cell area while minimizing PCE losses is another area which needs further development. The research-cell efficiencies in [1, Figure 1] show that the highest PCE single junction PSCs have areas < 1cm². Further research and development will need to go into understanding the PCE degradation associated with scaling up module area to sizes viable for commercialization. Progress has been made in understanding the PCE degradation which can partly be attributed to the non-uniform coating of the perovskite material [3]. As seen in [3, Figure 4], over the past few years perovskite cell efficiency has grown significantly when scaling small area cells into mini-modules.

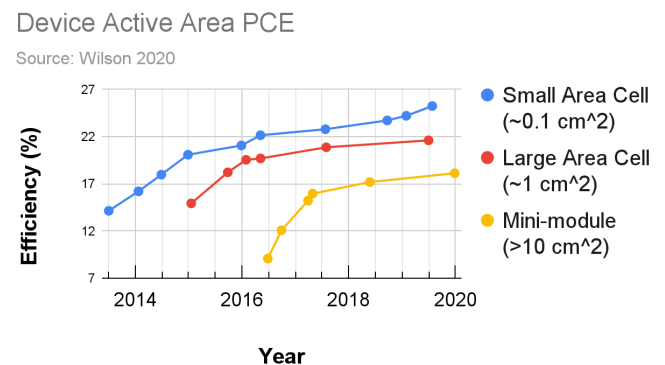


Figure 4. Device Active Area PCE

In addition to understanding technical benefits and limitations, it is important to understand the environmental and human impacts that perovskites have considering the majority of them contain Lead (Pb) as the organic cation in the molecular structure of the semiconductor. Although the amount of lead present in PSCs is small [4], wide adoption of the technology will require substantial efforts to understand and mitigate the risks that the Pb in PV has. According to [14], there is no known Pb exposure level that is safe, and exposure to the material has been shown to cause irreversible damage to humans. The use of this technology commercially will require thought into processes for recycling and disposing of these devices safely. Product life cycle assessments will be essential for Pb-based PSCs

to ensure that this new technology does not negatively impact society. Research into Pb-free perovskites is ongoing and presently none of these alternatives can match the PCE that Pb-based perovskites have [4]. The reliance on Pb for an efficient PV cell may be the biggest factor which will limit PSCs ability to reach commercial viability and should be the focus of future studies if Pb-based PSCs are deemed to be harmful to the public or environment.

VI. CONCLUSION

Perovskite-based photovoltaics are showing rapid improvements in their power conversion efficiency in a relatively short period when compared to other technologies. This novel cell technology will allow for new PV mounting opportunities by allowing installation on curved surfaces as well as in semi-transparent windows. Further research and testing are needed to address the instabilities of the technology. These instabilities cause degradation in the perovskite-based cell's PCE which will need to be mitigated to reach lifespans suitable for widespread commercial use. PSCs reliance on Pb to reach the efficiencies outlined in this paper is a concern, and more research will need to focus on how to limit the Pb from affecting humans and the environment. Alternatively, more research into lead-free perovskites which offer similar PCEs will need to be explored for this technology to be widely available to the public.

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